SMART DISPLAY MODULE SPECIFICATION

4.3 Inch Smart Display with TOUCH	
Model:	PLCM-UES31S043H800V480C-U
MCU：	ESP32-S31-WROOM-3
PCB：	UES31S043H800V480C-U
Display：	4.3" IPS, 800 × 480, ST7262, RGB666/888, GT911 CTP
Version:	V1.0
Date:	2026-9-3

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Customer Confirmation

Approved by	Notes

REVISION HISTORY

Revision	Date	Contents of Revision Change	Remark
V1.0	20260903		

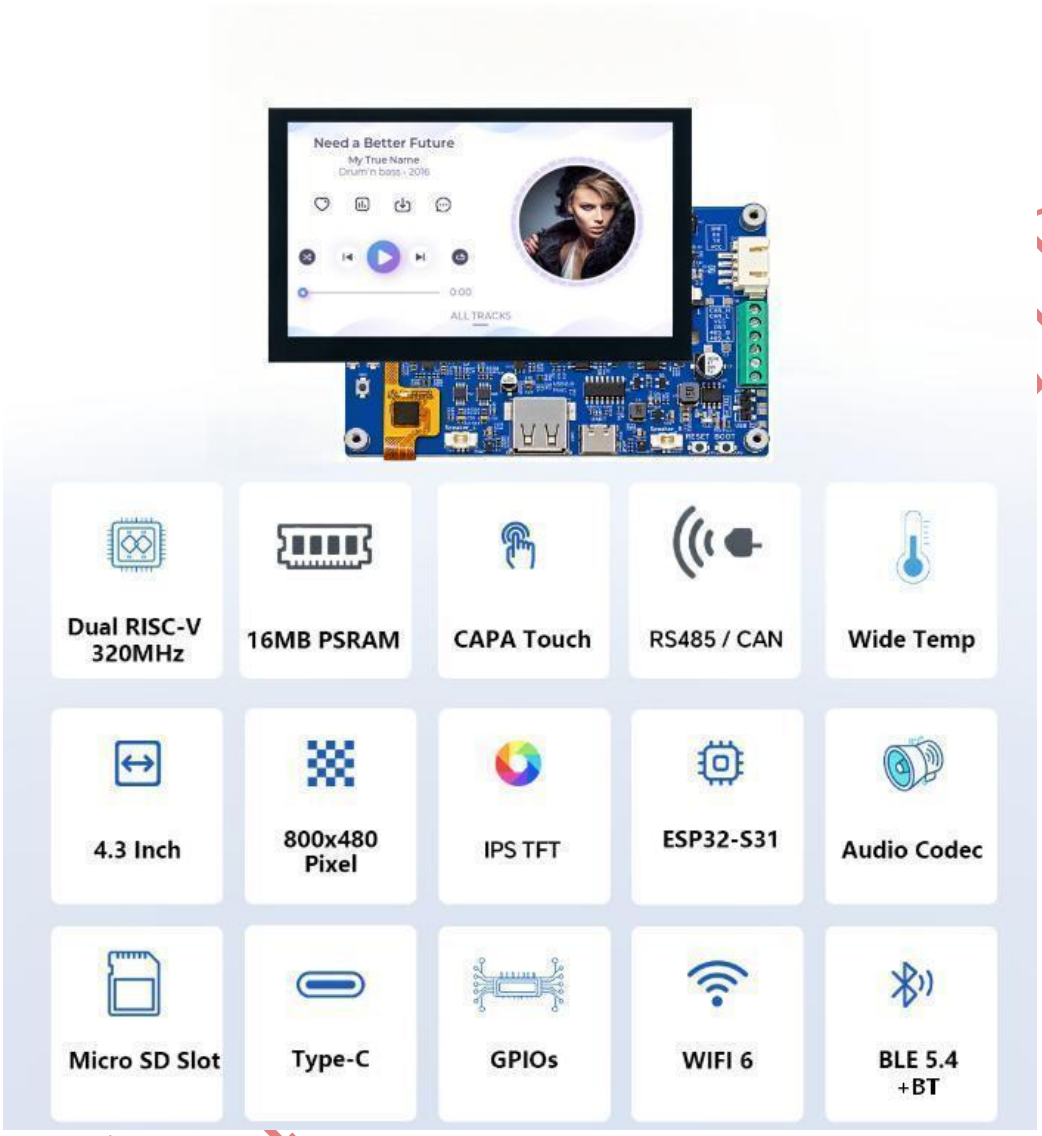
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1. Introduction



PLCM-UES31S043H800V480C-U is a high-performance smart display module designed by VIEWE. It is based on the Espressif ESP32-S31-WROOM-3 module and a 4.3-inch RGB capacitive touch panel (800 × 480). The LCD and touch devices are the same as those used on UEDX80480043E-WB-B (ST7262 driver IC and GT911 CTP). The main board is a new design around ESP32-S31: a dual-core 32-bit RISC-V MCU running at up to 320 MHz, with Wi-Fi 6, Bluetooth 5.4 (LE) plus Bluetooth Classic, USB 2.0 High-Speed OTG, stereo audio, SDIO 3.0 storage, and RS485 / CAN industrial interfaces.

Notice: ESP32-S31 is a preview device. ESP-IDF builds must use the --preview option (for example: idf.py --preview set-target esp32s31). Module ratings follow the Espressif ESP32-S31-WROOM-3 datasheet (pre-release). New designs that need USB 2.0 High-Speed, Wi-Fi 6, or on-board codec audio should use this S31 board rather than the ESP32-S3。

1.1 Product Features

CPU:

1. Processor
   1) ESP32-S31 RISC-V 32-bit dual-core processor, main frequency up to 320 MHz, plus a ULP-RISC-V coprocessor.
   2) 2.4 GHz Wi-Fi 6 (IEEE 802.11b/g/n/ax, HT20/40, up to 150 Mbps), Bluetooth 5.4 LE (including LE Audio), Bluetooth Classic (BR/EDR), and IEEE 802.15.4 (Zigbee / Thread).
   3) On-module PCB antenna.
   4) Hardware security: Secure Boot, Flash/PSRAM encryption, cryptographic acceleration, and TEE. See the chip datasheet for details.

2. Memory
   1) On-chip: 320 KB ROM, 512 KB SRAM, 32 KB low-power SRAM.
   2) Typical module configuration on this product: 16 MB Quad SPI Flash + 16 MB Octal SPI PSRAM (ESP32-S31-WROOM-3-N16R16V).
   3) Flash and PSRAM can be accessed in parallel.

3. Peripheral Interfaces
   1) USB Type-C port for 5 V power, firmware download, and serial debug (CH340C bridged to UART0).
   2) USB Type-A port connected to the dedicated USB 2.0 High-Speed OTG PHY (USB_DP / USB_DM, not GPIOs). Host 5 V is sourced through TPS2051C (~500 mA). VBUS-EN is hard-wired high and is not MCU-controlled.
   3) On-board MicroSD slot, 4-bit SDMMC / SDIO 3.0.
   4) ES8389 stereo codec, dual NS4150B speaker amplifiers, dual analog microphones, and L/R speaker connectors.
   5) RS485 (SIT3088E, hardware automatic direction) and CAN (SIT1050T + on-chip TWAI) on a 6-pin terminal.
   6) 2 × 10 pin 2.54 mm header (H1) bringing out UART0/UART1, I2C, RS485/CAN GPIOs, 3.3 V / 5 V / GND.
   7) Four ADC keys, RESET / BOOT, WS2812B RGB LED, and a 3 kHz passive buzzer（Not attached by default）.

   For more information on ESP32-S31-WROOM-3, please refer to the Espressif S31 datasheet .

Display

1) Size: 4.3 Inch
2) Resolution: 800 *480
3) Pixel Arrangement：RGB Vertical Stripe
4) Interface Mode：40PIN RGB 24bits
5) Driver IC： ST7262E43-G4

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6) Touch IC: GT911
7) Brightness: 400 cd/m^2
8) Touch: CTP
9) Verified timing: PCLK 18 MHz, negative polarity; HSYNC 1 / 40 / 20; VSYNC 1 / 10 / 5.

More information about Display can be found here:Display Specification.pdf

Other

1) Operation Temperature: −20 ~ 70 ° C (limited by the LCD).
2) Storage Temperature: −30 ~ 80 ° C.
3) MCU module rating: −40 ~ 85 ° C.

1.2 Applications

With rich connectivity and powerful processing capabilities, the UEDX80480043E-WB-B is an ideal choice for IoT devices in the following areas:

- Smart Home Control Panels
- Industrial Automation HMI
- Smart Appliances
- Consumer Electronics
- Wireless Data Loggers
- Touch Screen Interfaces
- Educational Learning Platforms

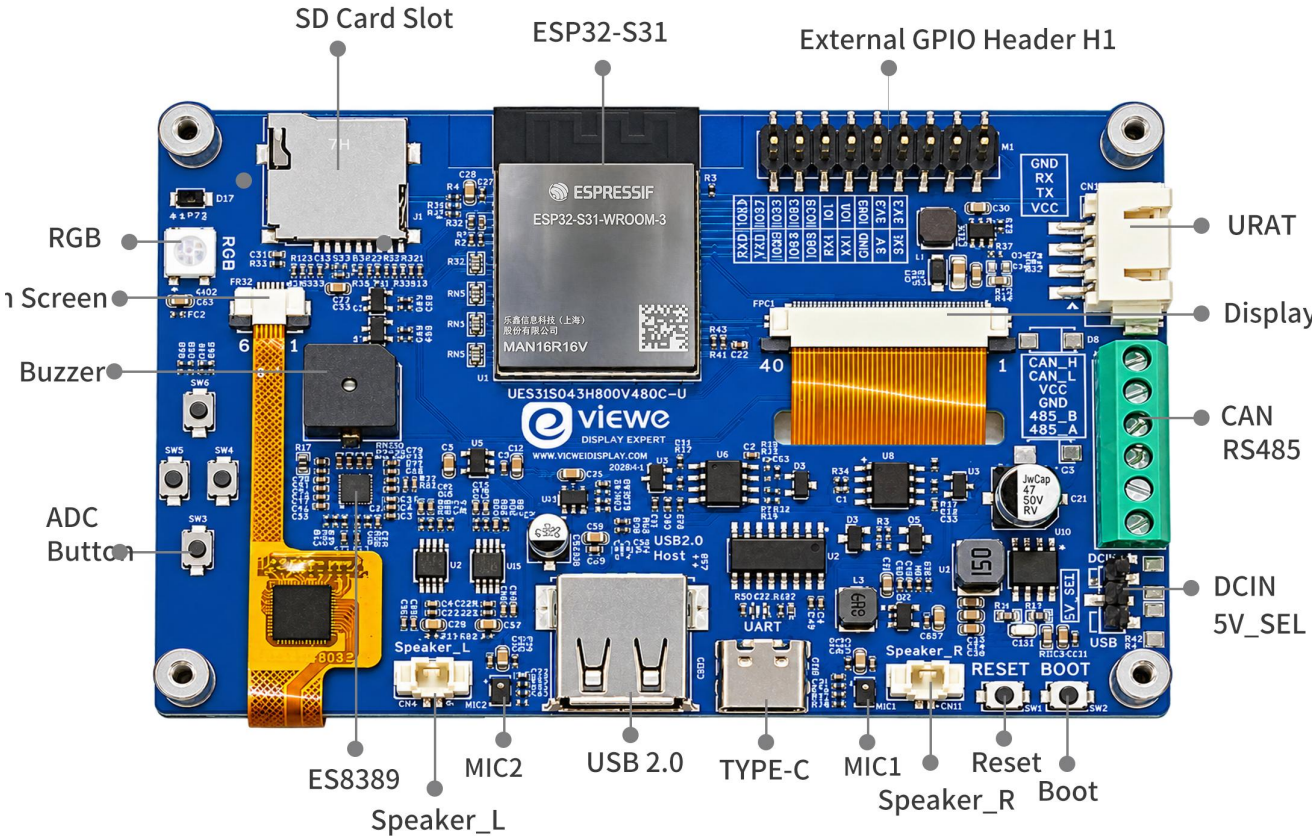
1.3 Product Naming

Field	Code	Meaning
Form	PLCM	PCB+LCM(Display)
Brand / series	UE	VIEWE smart display module
MCU	S31	ESP32-S31-WROOM-3
Size	S043	4.3 inch
Resolution	H800V480	800 × 480
Touch	C	Capacitive (GT911)
Bus	U	UART/RS485/I2C/CAN/USB 2.0 High-Speed OTG

2. Product information

2.1 Interface Description

This module achieves efficient communication with the main control unit via UART, RS485 and CAN buses, and supports USB 2.0 High-Speed OTG functions to meet diverse data interaction requirements. The capacitive touch screen adopts the GT911 solution to deliver smooth and responsive touch experience. The 4.3-inch screen features an 800×480 high-definition resolution, ensuring clear and delicate display content. Integrated with the ESP32-S31-WROOM-3, it balances high-performance processing and low-power operation. Suitable for various industrial and commercial application scenarios, it provides stable and reliable visual interaction support for devices.



1、Main Control Module:
ESP32-S31-WROOM-3 Dual-core RISC-V, up to 320 MHz, 16 MB Flash + 16 MB PSRAM, PCB antenna.

2、Display Interface:
40-pin RGB output. The panel is 24-bit capable; this board wires RGB666 (R2–R7 / G2–G7 / B2–B7). See the Display Interface table.

3、SD Card Slot：
SPI interface connected for external storage expansion, Please refer to Section 2.9 below for detailed information.

4、Touch interface：
I2C (SDA = GPIO0, SCL = GPIO1) to GT911. INT and RST are present on the 6-pin FPC; verified examples leave them unused (NC).

5、USB Type-C interface：
5 V DC input, programming, and serial debug through CH340C

6、USB Type-A:
USB 2.0 High-Speed OTG PHY (USB_DP / USB_DM). Host 5 V via TPS2051C. VBUS-EN is pulled high by hardware.Please refer to Section 2.7 below for detailed information.

7、UART helper (4-pin):
GND / RX / TX / VCC for an external serial adapter.The 4-pin pitch of 2.5 mm

8、RGB-LED (WS2812B)：
one WS2812B, DIN = GPIO37, powered from 5 V with a level-shift network.Please refer to Section 2.9 below for detailed information.

9、Boot Buttons：
BOOT button (GPIO61) for firmware download mode.

10、Reset Buttons：
RESET button (CHIP-EN) for resetting the board.

11、External GPIO header H1:
2 × 10 pins, 2.54 mm. Please refer to Section 2.6 below for detailed information.

12、Industrial terminal CN3 (CAN/RS485):
CAN_H, CAN_L, VCC, GND, 485_B, 485_A， Please refer to Section 2.8 below for detailed information.

13、Audio:
Dual- channel speaker interface Speaker L / R (CN2 / CN1) with a 2-pin pitch of 1.25 mm, and on - board dual microphones MIC1 / MIC2 deliver high-quality audio via ES8389 and NS4150B，Please refer to Section 2.5 below for detailed information.

14、ADC Button：
ADC key array SW3–SW6 on GPIO42. Please refer to Section 2.9 below for detailed information.

15、Buzzer:
BEEP_EN = GPIO46, high = on, 3 kHz passive type.Not attached by default

16、DCIN / 5V_SEL:
external 5 V input and 5 V source-selection jumper.

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Display Interface:

Pin No.	Symbol	I/O	Description
1	**LEDK**	P	Power supply for backlight cathode
2	**LEDA**	P	Power supply for backlight anode
3	**GND**	P	Power ground
4	**VDD**	P	Logic supply, 3.3 V
5-12	**R0-R7**	I	Red data. This board uses R2–R7 (GPIO2–7). R0/R1 = NC
13-20	**G0-G7**	I	Green data. This board uses G2–G7 (GPIO8–13). G0/G1 = NC
21-28	**B0-B7**	I	Blue data. This board uses B2–B7 (GPIO14–19). B0/B1 = NC
29	**GND**	P	Power ground
30	**CLK**	I	Pixel clock, negative polarity (GPIO40)
31	**DISP**	I	Standby mode. Normally pulled high
32	**HSYNC**	I	Horizontal sync, negative polarity (GPIO44)
33	**VSYNC**	I	Vertical sync, negative polarity (GPIO45)
34	**DEN**	I	Data enable. Display access is enabled when DE is “H” (GPIO43)
35	**NC**	I	Dummy
36	**GND**	P	Power ground
37	**XR**	-	Dummy
38	**YD**	-	Dummy
39	**XL**	-	Dummy
40	**YU**	-	Dummy

I：Input；O：Output；P：Power

TP Interface:

Pin No.	Symbol	I/O	Description
1	**RST**	P	Touch reset on FPC; not used
2	**3.3V**	P	3.3 V logic supply
3	**GND**	P	Power ground
4	**INT**	I	Touch interrupt on FPC;TP INT = GPIO38
5	**SDA**	I	SDA = GPIO0, shared with ES8389
6	**SCL**	P	SCL = GPIO1, 400 kHz, shared with ES8389

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2.2 GPIO Definition

GPIO Number	Current Usage	Function Description
CHIP-EN	Reset	RESET button (SW1)
GPIO0	I2C SDA	GT911 + ES8389. Pulled high at reset (normal boot). Not the BOOT key; also on H1
GPIO1	I2C SCL	GT911 + ES8389; also on H1
GPIO2–7	LCD-R2–R7	LCD red data, high 6 bits
GPIO8–13	LCD-G2–G7	LCD green data, high 6 bits
GPIO14–19	LCD-B2–B7	LCD blue data, high 6 bits
GPIO20–23	SD-D0–D3	SDMMC 4-bit data
GPIO24	SD-CLK	SDMMC clock
GPIO25	SD-CMD	SDMMC command
USB_DP / USB_DM	USB 2.0 HS PHY	Dedicated module pins. Do not configure as GPIO
GPIO33	UART1 TX	Verified as UART1 TX. Also a USB Serial/JTAG pad
GPIO34	UART1 RX	Verified as UART1 RX
GPIO35	RS485 TX	SIT3088E DI; S8050 auto-drives DE/~RE; also on H1
GPIO36	RS485 RX	SIT3088E RO; also on H1
GPIO37	WS2812 DIN	On-board XL-5050RGBC-WS2812B
GPIO39	LCD-BL-EN	Backlight enable, active high
GPIO40	LCD-PCLK	Pixel clock, 18 MHz, negative polarity
GPIO42	ADC KEY	SW3–SW6 resistor ladder, ADC1, ~0–2 V usable range
GPIO43	LCD-DE	Data enable
GPIO44	LCD-HS	Horizontal sync (GPIO number, not USB_DM)
GPIO45	LCD-VS	Vertical sync (GPIO number, not USB_DP)
GPIO46	BEEP-EN	Buzzer, active high; 3 kHz PWM recommended
GPIO47	PA-EN	NS4150B amplifier enable, active high
GPIO48	I2S MCLK	Present on schematic; firmware examples leave NC (codec no_mclk)
GPIO49	I2S BCK	ES8389 bit clock
GPIO50	I2S WS	ES8389 word select / LRCK
GPIO51	I2S DOUT	MCU → ES8389 DAC
GPIO52	I2S DIN	ES8389 ADC → MCU
GPIO53	CAN TX	SIT1050T TXD
GPIO54	CAN RX	SIT1050T RXD
GPIO55–57	H1 GPIO	Broken out on H1
TX0（IO58）/ RX0(IO59)	UART0	CH340C download / log; also on H1
GPIO60	SD-CTRL	SD 3.3 V power switch, active low
GPIO61	BOOT	BOOT button (SW2); also on H1

SD I2C Reset
RS485 UART1 USB2.0
Buzzer LCD RGB
CAN I2S PA-EN
GPIO UART0 Boot

2.3 Display Information

Item	Specification	Unit	Remark
Pixel Driving element	IPS TFT	-	-
Screen Size	4.3	Inch	Diagonal
Resolution	800(W)*3(RGB)*480(H)	Dots	-
Interface	RGB 24bits	-	40PIN
Module Power Consumption	1.04	Watt	Typ.
Active Area	95.04(W)*53.86(H)	mm	-
Pixel pitch (W*H)	0.1188(W)*0.1122(H)	mm	-
Module Size (W*H*D)	105.52(W)*67.17(H)*2.8(D)	mm	Tolerance:±0.2
Luminance	400	cd/m^2	Typ.
Viewing Direction	All	O'clock	-
Display Color	16.2M	Colors	24bits

2.4 LCD RGB timing (verified)

Parameter	Value
PCLK	18 MHz, pclk_active_neg = true
Horizontal resolution	800
Vertical resolution	480
HSYNC pulse / back porch / front porch	1 / 40 / 20
VSYNC pulse / back porch / front porch	1 / 10 / 5
Framebuffer	RGB888 in PSRAM, one frame buffer

2.5 Audio Interface

The audio codec is Everest ES8389 (I2C 7-bit address 0x20). I2S0 is used in full duplex. 48 kHz / 16-bit / stereo playback and record have been verified. MCLK is not required (codec configured with no_mclk). Dual NS4150B Class-D amplifiers drive the left and right speakers. PA_EN = GPIO47. MIC1 and MIC2 are on-board analog microphones.

Signal	GPIO / Device	Description
I2C SDA / SCL	GPIO0 / GPIO1	Shared with GT911, 400 kHz
I2S BCK	GPIO49	Bit clock
I2S WS	GPIO50	Word select / LRCK
I2S DOUT	GPIO51	DAC playback
I2S DIN	GPIO52	ADC record
I2S MCLK	NC in firmware	Optional on schematic
PA_EN	GPIO47	High = amplifier on
Speaker	CN1 / CN2	NS4150B differential output, typ. 4 Ω / 3 W class

2.6 H1 header (2 × 10, 2.54 mm), top to bottom

Number	Left	Number	Right
1	3V3	2	5V
3	3V3	4	5V
5	GND	6	GND
7	GPIO0 (SDA)	8	TX1 (GPIO33)
9	GPIO1 (SCL)	10	RX1 (GPIO34)
11	GPIO35 (485_TX)	12	GPIO36 (485_RX)
13	GPIO53 (CAN_TX)	14	GPIO54 (CAN_RX)
15	GPIO55	16	GPIO56
17	GPIO57	18	TX0
19	GPIO61 (BOOT)	20	RX0

GPIO35/36 and GPIO53/54 on H1 are in parallel with the on-board RS485 and CAN transceivers. Do not drive both the header and the terminal at the same time. GPIO0/1 already have I2C pull-ups.

2.7 USB 2.0 Interface

ESP32-S31 integrates a USB 2.0 High-Speed OTG PHY on dedicated USB_DP / USB_DM pins (package pins 44 / 45 of the chip). These are not GPIOs and need no GPIO-matrix setup. The board provides two physical connectors:

Port	Bridge / PHY	Role	Notes
Type-C	CH340C → UART0	Download / log / 5 V in	Hold BOOT (GPIO61), then tap RESET
Type-A	USB 2.0 HS PHY	OTG Host or Device	TPS2051C sources 5 V; VBUS-EN hard-wired high

Verified applications: (1) TinyUSB HID mouse mapped from the touch panel — class-compliant on Windows and Linux, no custom driver; (2) USB extended display — host-side IDD / vendor driver required, JPEG decode to the RGB panel. Type-A defaults to supplying 5 V as a USB Host. When the same port is used as a Device attached to a PC, avoid back-feeding VBUS from both ends.

2.8 UART / RS485 / CAN

Interface	Controller	MCU pins	External	Verified setting
UART0	CH340C	TX0 / RX0	Type-C / H1	Download and monitor
UART1	GPIO matrix	TX=33, RX=34	H1 TX1 / RX1	115200 8N1
RS485	SIT3088E + UART2	TX=35, RX=36	CN3 485_A / 485_B / GND	115200 8N1, auto DE
CAN	SIT1050T + TWAI	TX=53, RX=54	CN3 CAN_H / CAN_L / GND	500 kbit/s, Classic CAN

CN3 terminal (follow the PCB silkscreen)

Silk	Function
CAN_H	CAN high
CAN_L	CAN low
VCC	5 V — check jumper and load before using as a supply
GND	Common ground
485_B	RS485 B
485_A	RS485 A

RS485: DE/~RE is switched automatically from 485_TX through an S8050. Treat the port as a normal UART in software; do not configure RTS. 120 Ω termination and A-pull-up / B-pull-down are on board. CAN: 120 Ω termination on board; CAN_GND is tied to system ground through 0 Ω. Do not connect a USB-TTL adapter to A/B or CAN_H/L.

2.9 Storage, Keys, LED

Micro SD

SDMMC 4-bit: CLK = GPIO24, CMD = GPIO25, D0–D3 = GPIO20–23, SD_CTRL = GPIO60 (active low). Examples mount FAT at SDMMC_FREQ_HIGHSPEED. Assert GPIO60 low before initializing the host.

ADC keys (measured on the real board)

Key	Typical voltage	Software window	Remark
Idle	≈ 3.3 V (pull-up)	Saturated / idle	S31 ADC 0 dB ≈ 0–2 V; idle saturates
SW3	≈ 0.38 V	100–600 mV	GPIO42
SW4	≈ 0.82 V	600–1080 mV	GPIO42
SW5	≈ 1.34 V	1080–1605 mV	GPIO42
SW6	≈ 1.87 V	1605–2200 mV	GPIO42

Two keys cannot be detected at once (inherent to the divider). Silkscreen order on some drawings does not match the measured voltages; use the table above.

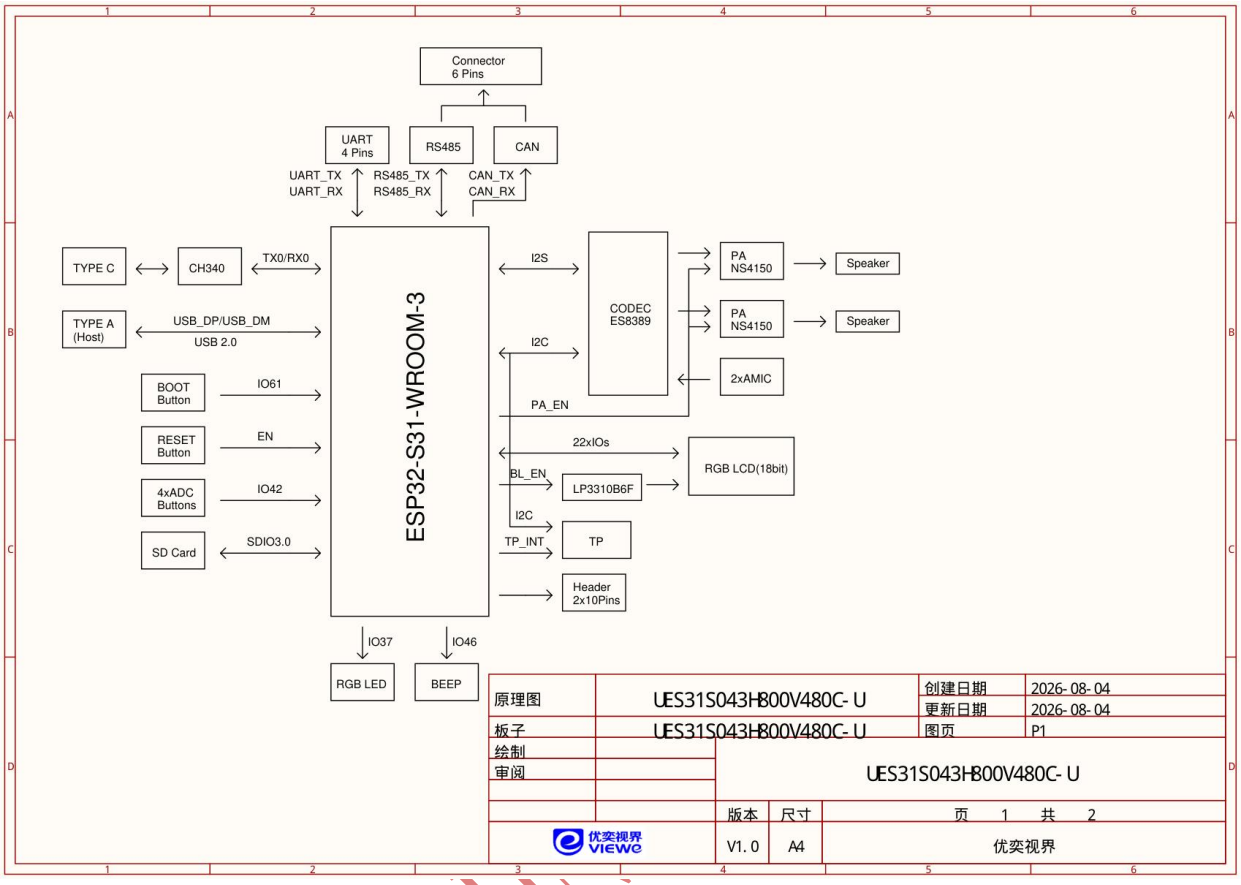
2.10 Voltage & Current

Item	Conditions	Min	Typ	Max	Unit
Power Voltage	DC	-	5.0	-	V
Operation Current	VCC= +5V, Maximum backlight current	80	320	500	mA
	VCC= +5V,backlight off	-	100	-	mA
Recommended power supply:5V 1A DC					

2.11 Reliability Test

Item	Conditions	Min	Typ	Max	Unit
Working Temperature	60%RH at 5V voltage	-20	25	70	C
Storage Temperature	---	-30	25	85	C
Working Humidity	25°C	10%	60%	90%	RH
ESD	---	Contact: ±4KV Air: ±8KV			KV

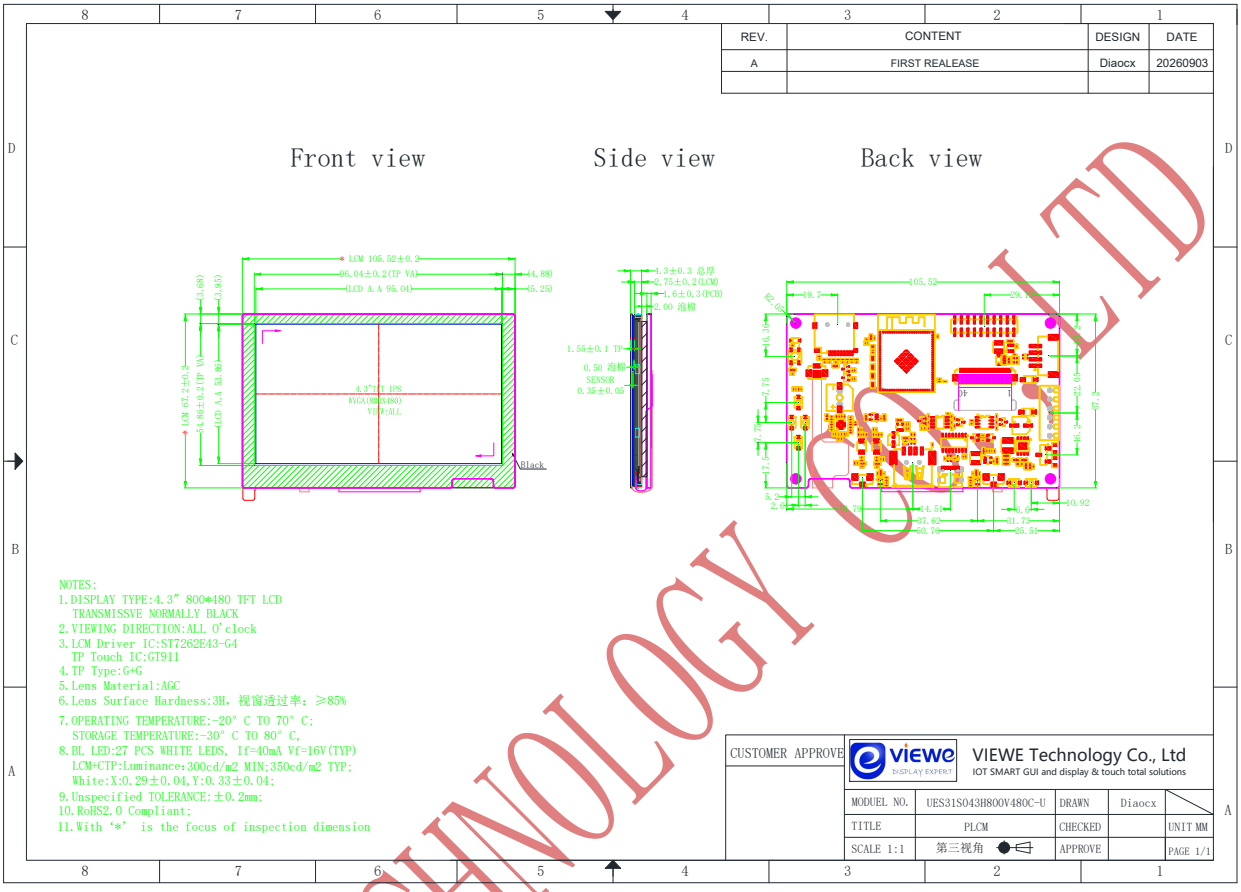
3. Functional Block Diagram



Note: ESP32-S31 includes a 2.4 GHz radio that supports Wi-Fi 6, Bluetooth 5.4 (LE), Bluetooth Classic, and 802.15.4. Because they share the same RF front-end, Wi-Fi and Bluetooth cannot transmit or receive simultaneously; the radio switches between protocols as needed. USB 2.0 High-Speed is an independent PHY and does not share that front-end.

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4. MECHANICAL DRAWING



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5. Detailed information (codes, drawings...)

Github： https://github.com/VIEWESMART/PLCM-UES31S043H800V480C-U

Gitee: https://gitee.com/VIEWESMART/PLCM-UES31S043H800V480C-U

Wiki： https://displaywiki.com/products/esp32-s3-4.3inch-800x480-touch-uart-display-hmi

Official Store：

https://viewedisplay.com/product/4-3-inch-800x480-esp32-s31-wifi6-touch-smart-hmi-uart-display

Prompt: The materials are being developed and enriched. Please stay tuned.

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